Clémence MÉTAYER

4th year PhD Student | Applied Mathematics in Cancer Science

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23 rue Dugommier, 75012 Paris, France
Born on May 15, 1996 (29 years old) in Angers, France

As a PhD student in applied mathematics for cancer biology, I develop and study model learning methods for discovering and identifying dynamic systems from experimental data. My work focuses in particular on the circadian analysis of omics data and the inference of parsimonious differential models, with applications to the circadian rhythm, the immune system, and the optimization of cancer treatments.

PROFESSIONAL EXPERIENCE

Now – October 2022
PhD student in applied mathematics in cancer science, CURIE INSTITUTE, INSERM, INRIA, Paris, France
Supervision : Dr. Annabelle Ballesta, Dr. Anne-Laure Huber, Dr. Samuel Bernard

- Development and study of model learning methods for automatically discovering EDOs systems from data
- Estimation of EDOs model parameters
- Development of a pipeline for rhythmic analysis of omics data (modeling, statistics, differential analysis, enrichment)
- Acquisition of knowledge about cancer, the circadian clock, the immune system, and DNA damage repair
- Introduction to experimental procedures (cell culture, synchronization, siRNA transfections, and cell collection)
- Writing scientific articles (literature review, methodology paper, application paper)
- Scientific monitoring

September 2022 – March 2022
Mathematics tutor for students in MP, MPSI, and PCSI preparatory classes, OZANAM CENTER, Lyon

- Hosting group and individual mentoring sessions
- Answering students' questions and clarifying complex points in the course

October 2021 – April 2021
Research Intern in Biostatistics, INRIA SISTM, Bordeaux, France
Supervision : Dr. Marta Avalos Fernandez

- Systematic literature review and meta-analysis of data from several studies
- Implementation of analyses and visualizations
- Acquisition of knowledge about chronic respiratory diseases, the respiratory microbiota, and the different types of diversity within this microbiota
- Writing a research article presenting the results
- Scientific monitoring

June 2020 – May 2020
Research Intern in Epidemiology, INRIA DRACULA, Lyon, France
Supervision : Dr. Mostafa Adimy

- Modeling of secondary dengue virus infection using a system of EDOs
- Mathematical analysis of this system (reproduction rate R0, equilibria, stability of equilibria)
- Acquisition of knowledge about the dengue virus and the phenomenon of antibody-dependent enhancement of infection
- Writing a research article presenting the results
- Scientific monitoring

Formation

Academic

2019 - 2021 **Master's degree in Applied Mathematics and Statistics, major in Biology and Medicine**, Claude Bernard Lyon 1 University, France
2016 - 2019 **Bachelor's degree in General Mathematics and Applications**, Claude Bernard Lyon 1 University, France
2014 - 2016 **First Year Common to Health Studies (Medical school)**, Angers University, France
2014 **Scientific Baccalauréat Diploma with mathematical major**, Saint-Joseph La Pommeraye High School, France

Other training courses

2025 **Machine Learning Specialization** (supervised machine learning, advanced learning algorithms, unsupervised learning, reinforcement learning), Stanford University & DeepLearning.AI
2025 **International Course on Molecular and Mechanical Oscillations**, Curie Institute
2024 **Next Generation Sequencing & Cancer**, Canceropôle Île-de-France
2023 **Introduction to Bayesian Data Analysis**, openHPI

Research publications

1) **Métayer, C.**, Ballesta, A., & Martinelli, J. (2026). Data-driven discovery of digital twins in biomedical research. Briefings in Bioinformatics, 27(1), bbaf722.
2) Avalos-Fernandez, M., Alin, T.*, **Métayer, C.***, Thiébaut, R., Enaud, R., & Delhaes, L. (2022). The respiratory microbiota alpha-diversity in chronic lung diseases : first systematic review and meta-analysis. Respiratory Research, 23(1), 214.
3) de A. Camargo, F., Adimy, M., Esteva, L., **Métayer, C.**, & Ferreira, C. P. (2021). Modeling the Relationship Between Antibody-Dependent Enhancement and Disease Severity in Secondary Dengue Infection. Bulletin of Mathematical Biology, 83(8), 85.

- **Métayer, C.**, Bardoulet, L., Petrilli, V., Martinelli, J., Bernard, S., Huber A.-L., & Ballesta, A. Discovering the Dynamical Interactions between the Circadian Clock, the NLRP3 Immune Receptor, and ATM-Mediated DNA Damage Response. (en préparation)
- **Métayer, C.**, Kassem, S., Martinelli, J., & Ballesta, A. BaSIC : Bayesian Sparse Identification of Nonlinear Dynamics for Multi-Conditions Data. (en préparation)
- **Métayer, C.***, Tuffet, R.*, Fawaz, R., Robert, R., Aubin, J.-B., & Duraffourg, M. Predictive Factors of Surgical Success in Pudendal and Inferior Cluneal Neuralgia Caused by Entrapment Syndromes. (en rédaction)

Conferences and Seminars

Oral presentations

July 2026 **Society for Mathematical Biology Annual Meeting** (Graz, Autriche) — Mini-symposium talk (invited) : *Learning dynamical models of interactions between the NLRP3 immune receptor and the circadian clock : optimizing lung cancer treatments*.
July 2024 **Society for Mathematical Biology Annual Meeting** (Séoul, Corée du Sud) — Contributed talk : *Learning dynamical models of interactions between the NLRP3 immune receptor and the circadian clock : optimizing lung cancer treatments*.

Posters

October 2023 **48th Congress of the Francophone Society of Chronobiology** (Paris) — Poster : *Learning dynamical models of the interactions between the immune receptor NLRP3 and the circadian clock – application to lung cancer*.

September 2023	**Workshop on Mathematical Perspectives on Immunobiology** (Blagoevgrad, Bulgarie) — Poster : *Learning dynamical models of the interactions between the immune receptor NLRP3 and the circadian clock – application to lung cancer*.
July 2023	**Society for Mathematical Biology Annual Meeting** (Columbus, Ohio, USA) — Poster : *Learning dynamical models of the interactions between the immune receptor NLRP3 and the circadian clock – application to lung cancer*.

Participation

April 2026	**Personalized Cancer Chronotherapy Conference** (Chicago University, Paris center)
May 2025	**Workshop "Towards Personalized Cancer Chronotherapy"** (Chicago University, Paris center)
May 2024	**Workshop "Towards Personalized Cancer Chronotherapy"** (Chicago University, Paris center)
April 2024	**Workshop "Mathematical and numerical tools for Oncology"** (Oncolille Institue, Lille, France)
October 2023	**Workshop "MC2D : Mathematical Challenges in Modelling Cancer Dynamics"** (Sorbonne Université, Paris)
April 2023	**Workshop "Towards Personalized Cancer Chronotherapy"** (Chicago University, Paris center)

Outreach in sciences

January 2026	**"Girls and Maths"** - Presentation of mathematical modeling applied to biology and health to high school students
October 2025	**Fête de la Science** — Animation of a mathematical modelling session applied to the personalisation of cancer treatments
October 2023	**Fête de la Science** — Animation of a mathematical modelling session applied to the personalisation of cancer treatments

Supervision

May - August 2026	**Stefan Kassem - Co-supervision of Master's 1 internship**, Comparison of model learning methods on noisy, missing, and multi-condition data

Awards and Grants

July 2023	**Society for Mathematical Biology Travel Grant**, Workshop on Mathematical Perspectives on Immunobiology, Blagoevgrad, Bulgarie

Skills

Programming	**Python** (NumPy, SciPy, Pandas, Matplotlib, Seaborn, scikit-learn, PyTorch, JAX, SymPy, Statsmodels, Shiny for Python), **R**, MATLAB
Vizualisation & illustrations	Adobe Illustrator, Inkscape
Development	Git, GitHub/GitLab, Visual Studio Code, Spyder, RStudio, Jupyter Notebook
Scientific writing	LaTeX, Overleaf, Microsoft Word, Google Docs
Operating systems	macOS, Windows

Expertises

Article review Mathematical Biosciences

Languages

French ●●●●●

English ●●●●○

Références

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